

Data Vortex — Round 1, Phase 2

Analytical Core Rebuild: SQL Reasoning Challenges
Participant: Sahil Khetal | Tool: SQLite 3 | Tables: users (1,500 rows), posts (12,000 cleaned rows), posts_raw (12,360 original rows, used only for H5)

Key Insights Summary:

- Facebook has the most posts (2,074), but Instagram has the highest average engagement per post (4,041.13)
- **Follower count does NOT predict engagement** — high-follower (4,002.01 avg) and low-follower (4,006.55 avg) users perform nearly identically (M2)
- No "super-user" or "viral post" outliers exist in this dataset — H1 and H3 both return 0 rows even at a 2x-average threshold, confirming engagement is unusually uniform
- 94 small-audience users (<10,000 followers) show suspiciously high, above-average engagement with unusual share patterns (H6)
- 4,536 rows in the original corrupted dataset carried data quality issues, all resolved in Phase 1 cleaning (H5)

EASY LEVEL

E1 — Platform Popularity

```
SELECT platform, COUNT(*) AS post_count
FROM posts
WHERE platform IS NOT NULL AND platform != 'Unknown'
GROUP BY platform
ORDER BY post_count DESC
LIMIT 1;
```

platform	post_count
Facebook	2074

Logic: Filtered out NULL and 'Unknown' platform values (data lost during corruption), grouped remaining rows by platform, counted posts, and took the single highest count.

E2 — Most Engaged Posts

```
SELECT post_id, platform, likes, shares, comments,
       (likes + shares + comments) AS total_engagement
FROM posts
WHERE likes IS NOT NULL
ORDER BY total_engagement DESC
LIMIT 10;
```

post_id	platform	likes	shares	comments	total_engagement
ycjj5zzt7mvx	Instagram	4983	1919	991	7893
wo7py9aljg3t	Reddit	4864	1981	948	7793
gmoeib832zbs	Facebook	4902	1880	982	7764
5kvuyvf38nqx	YouTube	4923	1971	861	7755
pvgl3d8hj7jd	Instagram	4989	1840	909	7738
tdgjyypua20	Unknown	4979	1932	812	7723
tne7s3o4l4wd	Instagram	4931	1903	878	7712
a1kiwl618kzy	Facebook	4811	1952	920	7683
fp89q1ickn9w	Twitter	4740	1933	955	7628
5n161ir5hhhr	YouTube	4751	1981	878	7610

Logic: Computed a derived 'total_engagement' column as the sum of likes, shares and comments, then sorted descending and took the top 10 rows.

E3 — Average Engagement by Platform

```
SELECT platform,
       ROUND(AVG(likes),2) AS avg_likes,
       ROUND(AVG(shares),2) AS avg_shares,
       ROUND(AVG(comments),2) AS avg_comments,
       ROUND(AVG(likes+shares+comments),2) AS avg_total_engagement
FROM posts
WHERE platform != 'Unknown'
GROUP BY platform
ORDER BY avg_total_engagement DESC;
```

platform	avg_likes	avg_shares	avg_comments	avg_total_engagement
Instagram	2500.49	1040.84	499.8	4041.13
YouTube	2514.66	1011.83	504.38	4030.87
Facebook	2524.11	984.17	506.94	4015.22
Reddit	2489.52	1002.23	511.18	4002.93
Twitter	2447.23	1005.39	506.13	3958.74

Logic: Grouped by platform and applied AVG() to each engagement metric independently, plus a combined average. Sorted to reveal Instagram as the highest average-engagement platform.

E4 — Highly Shared but Poorly Liked

```
SELECT post_id, platform, likes, shares, comments
FROM posts
WHERE shares > 1500 AND likes < 500
ORDER BY shares DESC;
```

Showing 10 of 247 total matching rows.

post_id	platform	likes	shares	comments
euvr0r10wrj6	Facebook	453	2000	408
oiszojqm6qnn	Instagram	390	1999	858
f2e5kdfldedz	Unknown	447	1997	641
mgv7p46wzpek	Reddit	252	1993	971
u1aa801qvxeu	Unknown	162	1992	306
6s9mxoemn488	Twitter	169	1991	856
0nsga7zrxpvt	YouTube	390	1991	848
lit2hyqg0v0l	Facebook	14	1990	769
0xytg6ityrle	Instagram	322	1989	170
twgx52qb72eo	YouTube	462	1989	636

Logic: Simple two-condition filter (shares > 1500 AND likes < 500) directly on the cleaned table. 247 posts matched; showing the top 10 by share count.

E5 — Users With Large Audiences

```
SELECT user_id, location, language, follower_count
FROM users
WHERE follower_count > 40000
ORDER BY follower_count DESC;
```

Showing 10 of 293 total matching rows.

user_id	location	language	follower_count
user_3o7w66o2	Berlin, Germany	ja	49944
user_u98jwp3f	Chicago, USA	zh	49936
user_usts5yuo	Shanghai, China	ja	49933
user_d4eat3v3	Tokyo, Japan	de	49914
user_siuvpkza	Chicago, USA	en	49905
user_ujllj1n7	Chicago, USA	es	49855
user_kr0ydzhh	Barcelona, Spain	fr	49836
user_9asov0m8	Singapore	hi	49763
user_84k7x6zn	Barcelona, Spain	ja	49736
user_t028mbub	Osaka, Japan	en	49727

Logic: Direct filter on the users table for follower_count > 40,000. 293 users matched; showing the top 10 by follower count.

MEDIUM LEVEL

M1 — Which Locations Generate the Most Engagement?

```
SELECT u.location, COUNT(p.post_id) AS number_of_posts,
       SUM(p.likes + p.shares + p.comments) AS total_engagement
FROM posts p
JOIN users u ON p.user_id = u.user_id
GROUP BY u.location
ORDER BY total_engagement DESC;
```

Showing 10 of 33 total matching rows.

location	number_of_posts	total_engagement
Los Angeles, USA	459	1838780
Munich, Germany	452	1812255
Shanghai, China	451	1796029
Barcelona, Spain	439	1788194
Melbourne, Australia	421	1716188
Dubai, UAE	421	1706870
Houston, USA	422	1669211
Osaka, Japan	398	1640035
Rio de Janeiro, Brazil	414	1629468
Mumbai, India	413	1613909

Logic: Joined posts to users on user_id to bring location into the posts context, then grouped by location and summed total engagement. 33 unique locations found; Los Angeles ranks #1.

M2 — Do High Follower Users Get More Engagement?

```
SELECT CASE WHEN u.follower_count >= 25000 THEN 'High (>=25000)'
           ELSE 'Low (<25000)' END AS follower_group,
       COUNT(p.post_id) AS post_count,
       ROUND(AVG(p.likes + p.shares + p.comments), 2) AS avg_engagement_per_post
FROM posts p
JOIN users u ON p.user_id = u.user_id
GROUP BY follower_group;
```

follower_group	post_count	avg_engagement_per_post
High (>=25000)	5925	4002.01
Low (<25000)	6075	4006.55

Logic: Used a CASE expression to bucket users into two follower groups, then compared average engagement per post between groups. KEY INSIGHT: 4002.01 vs 4006.55 -- essentially no difference. Follower count does not predict engagement in this dataset.

M3 — Most Active Users

```
SELECT p.user_id, u.follower_count, u.location,
       COUNT(p.post_id) AS post_count,
       SUM(p.likes + p.shares + p.comments) AS total_engagement
FROM posts p
JOIN users u ON p.user_id = u.user_id
GROUP BY p.user_id
ORDER BY post_count DESC
LIMIT 10;
```

user_id	follower_count	location	post_count	total_engagement
user_zqv2vrf5	13531	Vancouver, Canada	22	90718
user_nfo3ih5u	40429	Barcelona, Spain	19	76525
user_q3vs0uj7	32714	Manchester, UK	18	68430
user_n0ok02rt	2531	Dubai, UAE	18	62901
user_8wk0j5ae	36943	Los Angeles, USA	18	64619
user_0irp4abu	44206	Toronto, Canada	17	74461
user_xetn4exw	14902	Dubai, UAE	16	62175
user_wyeb78qv	24634	Dubai, UAE	16	55214
user_ujllj1n7	49855	Chicago, USA	16	71576
user_uerv85na	1824	Rome, Italy	16	72247

Logic: Grouped posts by user_id, counted posts per user and summed their engagement, joined to users for context, ranked by post_count.

M4 — Platform Behaviour by High Follower Users

```
SELECT p.platform, COUNT(p.post_id) AS post_count,
       ROUND(AVG(p.likes + p.shares + p.comments), 2) AS avg_engagement
FROM posts p
JOIN users u ON p.user_id = u.user_id
WHERE u.follower_count >= 30000 AND p.platform != 'Unknown'
GROUP BY p.platform
ORDER BY avg_engagement DESC;
```

platform	post_count	avg_engagement
Instagram	809	4129.89
Facebook	827	3992.78
Twitter	785	3986.03
Reddit	799	3979.17
YouTube	846	3961.53

Logic: Filtered to only posts by users with >=30,000 followers, then grouped by platform. Instagram gives this high-follower segment the best average engagement (4129.89).

M5 — Detect Suspicious Engagement

```
SELECT post_id, platform, likes, shares, comments,
       (likes + comments) AS likes_plus_comments
FROM posts
WHERE shares > (likes + comments)
ORDER BY shares DESC
LIMIT 20;
```

Showing 10 of 20 total matching rows.

post_id	platform	likes	shares	comments	likes_plus_comments
euvr0r10wrj6	Facebook	453	2000	408	861
oiszojqm6qnn	Instagram	390	1999	858	1248
2xcg9ld7du67	Twitter	1425	1999	440	1865
qq86lkjrfzlt	YouTube	897	1999	850	1747
sjv1fkjkr8e1	Unknown	1524	1998	129	1653
rrxmp0nra55	Instagram	981	1997	958	1939
zvj4ja8bp4xx	Unknown	1503	1997	338	1841
f2e5kdfldedz	Unknown	447	1997	641	1088
x8wq022t0pa5	Unknown	1380	1997	201	1581
lx50tyodyt6m	Instagram	1545	1996	63	1608

Logic: Compared shares directly against the sum of likes+comments in the WHERE clause -- posts where sharing behaviour outweighs on-platform reactions combined, which can indicate bot activity or external virality.

HARD LEVEL

H1 — Users With Abnormally High Engagement

```
WITH user_engagement AS (
  SELECT p.user_id, COUNT(p.post_id) AS post_count,
         AVG(p.likes+p.shares+p.comments) AS avg_engagement
  FROM posts p GROUP BY p.user_id
),
overall_avg AS (
  SELECT AVG(likes+shares+comments) AS overall_avg_engagement FROM posts
)
SELECT ue.user_id, u.location, u.follower_count, ue.post_count,
       ROUND(ue.avg_engagement,2) AS avg_engagement_per_post
FROM user_engagement ue
JOIN users u ON ue.user_id = u.user_id
CROSS JOIN overall_avg oa
WHERE ue.avg_engagement > 2 * oa.overall_avg_engagement
ORDER BY avg_engagement_per_post DESC;
```

Query returned 0 rows (see explanation below — this is a verified, genuine finding).

Logic: Two CTEs: one computing each user's average engagement, one computing the dataset-wide average. Compared per-user average against 2x the overall average ($4004.31 \times 2 = 8008.61$). RESULT: 0 rows -- verified genuine, since the highest individual user average is only ~5898.33. This shows engagement is uniformly distributed with no 'super-user' outliers.

H2 — Rank Users Within Their Location

```
WITH user_totals AS (  
  SELECT u.location, p.user_id,  
         SUM(p.likes+p.shares+p.comments) AS total_engagement  
  FROM posts p JOIN users u ON p.user_id=u.user_id  
  GROUP BY u.location, p.user_id  
)  
ranked AS (  
  SELECT location, user_id, total_engagement,  
         RANK() OVER (PARTITION BY location ORDER BY total_engagement DESC) AS rnk  
  FROM user_totals  
)  
SELECT location, user_id, total_engagement, rnk  
FROM ranked WHERE rnk <= 3  
ORDER BY location, rnk;
```

Showing 9 of 99 total matching rows.

location	user_id	total_engagement	rnk
Barcelona, Spain	user_nfo3ih5u	76525	1
Barcelona, Spain	user_5s9ifp0y	59036	2
Barcelona, Spain	user_24wzfb8b	58035	3
Beijing, China	user_3jedn9am	51317	1
Beijing, China	user_ttlouvlq	49742	2
Beijing, China	user_cdzp4vm2	48019	3
Berlin, Germany	user_m5z8a3sg	58947	1
Berlin, Germany	user_6e99lerx	56761	2
Berlin, Germany	user_rtp2dykx	52748	3

Logic: Used the RANK() window function PARTITIONed BY location, ordered by each user's total engagement -- this ranks users independently within each location, then filtered to keep only rank 1-3. 99 rows = 33 locations x top 3.

H3 — Platform Performance Compared With Its Own Average

```
WITH platform_avg AS (  
  SELECT platform, AVG(likes+shares+comments) AS avg_eng  
  FROM posts WHERE platform != 'Unknown' GROUP BY platform  
)  
SELECT p.post_id, p.platform,  
       (p.likes+p.shares+p.comments) AS post_engagement,  
       ROUND(pa.avg_eng,2) AS platform_avg_engagement  
FROM posts p  
JOIN platform_avg pa ON p.platform = pa.platform  
WHERE (p.likes+p.shares+p.comments) >= 2 * pa.avg_eng  
ORDER BY post_engagement DESC;
```

Query returned 0 rows (see explanation below — this is a verified, genuine finding).

Logic: Computed each platform's own average engagement in a CTE, joined back to posts, and flagged posts at least 2x their platform's average. RESULT: 0 rows -- verified genuine. The highest single-post engagement (7,893) falls short of even Twitter's threshold (~7,917, the lowest of all 5 platforms). No viral outliers exist in this dataset at the post level either.

H4 — Follower to Engagement Anomaly

```
WITH user_totals AS (  
  SELECT p.user_id, SUM(p.likes+p.shares+p.comments) AS total_engagement  
  FROM posts p GROUP BY p.user_id  
)  
,  
ranked AS (  
  SELECT user_id, total_engagement,  
         PERCENT_RANK() OVER (ORDER BY total_engagement DESC) AS pct_rank  
  FROM user_totals  
)  
SELECT r.user_id, u.follower_count, u.location, r.total_engagement  
FROM ranked r JOIN users u ON r.user_id = u.user_id  
WHERE u.follower_count < 5000 AND r.pct_rank <= 0.10  
ORDER BY r.total_engagement DESC;
```

Showing 10 of 17 total matching rows.

user_id	follower_count	location	total_engagement
user_uerv85na	1824	Rome, Italy	72247
user_hdas0iau	1620	Rio de Janeiro, Brazil	64069
user_n0ok02rt	2531	Dubai, UAE	62901
user_fgjkkrie	2211	Lyon, France	62690
user_6wra58f7	4953	Johannesburg, South Africa	57389
user_5oe5t3js	3151	London, UK	55379
user_rr1uzkql	1069	Milan, Italy	54561
user_r7eg1rac	2052	Houston, USA	54552
user_67hyf45u	898	Vancouver, Canada	52158
user_ogtvuuki	4459	Cairo, Egypt	52060

Logic: Used PERCENT_RANK() window function to compute each user's percentile rank by total engagement across the ENTIRE user base, then filtered for low-follower users (<5,000) still landing in the top 10% overall. 17 users found -- small accounts with disproportionately high total engagement, likely driven by high posting frequency.

H5 — Identify Data Anomalies (uses posts_raw)

```
SELECT post_id,  
       CASE  
         WHEN likes < 0 THEN 'Negative Likes'  
         WHEN platform IS NULL THEN 'Missing Platform'  
         WHEN text_content IS NULL THEN 'Missing Text Content'  
         WHEN text_content LIKE '%&%' OR text_content LIKE '%<%'  
           OR text_content LIKE '%>%' OR text_content LIKE '%  
%'  
           OR text_content LIKE '%  
%' THEN 'HTML Entities/Tags in Text'  
         ELSE 'Other'  
       END AS anomaly_type  
FROM posts_raw  
WHERE likes < 0 OR platform IS NULL OR text_content IS NULL  
      OR text_content LIKE '%&%' OR text_content LIKE '%<%'  
      OR text_content LIKE '%>%' OR text_content LIKE '%  
%'  
      OR text_content LIKE '%  
%'  
ORDER BY anomaly_type;
```

Showing 4 of 4536 total matching rows.

anomaly_type	count
Missing Platform	1752
Missing Text Content	1439
HTML Entities/Tags in Text	820
Negative Likes	525

Logic: IMPORTANT: run against posts_raw (the original corrupted table), not the cleaned posts table -- Phase 1 already fixed these issues, so querying the cleaned table would return 0 anomalies. A CASE expression classifies each flagged row into one anomaly type (first match wins for rows with multiple issues). 4,536 total anomalous rows found across 4 categories (summary shown; full 4536-row detail available in the GitHub repo query output).

H6 — Most Suspicious High Impact Users

```
WITH user_stats AS (
  SELECT p.user_id, COUNT(p.post_id) AS post_count,
         AVG(p.likes+p.shares+p.comments) AS avg_engagement,
         SUM(p.likes+p.shares+p.comments) AS total_engagement,
         MAX(CASE WHEN p.shares > p.likes THEN 1 ELSE 0 END) AS has_shares_gt_likes
  FROM posts p GROUP BY p.user_id
),
overall_avg AS (
  SELECT AVG(likes+shares+comments) AS overall_avg_engagement FROM posts
)
SELECT us.user_id, u.location, u.follower_count, us.post_count,
       ROUND(us.avg_engagement,2) AS avg_engagement, us.total_engagement
FROM user_stats us
JOIN users u ON us.user_id = u.user_id
CROSS JOIN overall_avg oa
WHERE u.follower_count < 10000
      AND us.avg_engagement > oa.overall_avg_engagement
      AND us.has_shares_gt_likes = 1
ORDER BY us.total_engagement DESC;
```

Showing 10 of 94 total matching rows.

user_id	location	follower_count	post_count	avg_engagement	total_engagement
user_uerv85na	Rome, Italy	1824	16	4515.44	72247
user_68enpikx	Houston, USA	9364	15	4520.53	67808
user_hdas0iau	Rio de Janeiro, Brazil	1620	16	4004.31	64069
user_fgjkkrie	Lyon, France	2211	14	4477.86	62690
user_9mtets0p	Johannesburg, South Africa	8262	14	4458.14	62414
user_87kc1g6d	Houston, USA	7238	14	4123.21	57725
user_g9g7onlw	Munich, Germany	7981	13	4422.69	57495
user_cssz1rea	Tokyo, Japan	7863	14	4105.0	57470
user_6wra58f7	Johannesburg, South Africa	4953	14	4099.21	57389
user_14ibkds2	Singapore	7361	13	4388.0	57044

Logic: Combined three conditions in a single CTE: low followers (<10,000), above-average engagement, AND at least one post with shares > likes (via a MAX/CASE flag). All three must be true simultaneously. 94 users matched, ranked by total engagement to surface the most 'suspicious' (high-impact-despite-small-audience) accounts first.